

Six Influential Muslim Scientists

1. Hasan Ibn Al-Haytham (965 - 1040)

The Iraqi pioneer, who is considered the first experimental physicist, established the modern scientific method more than 150 years before European scientists came to know it. He was considered as the Father of Modern Optics.

Haytham wrote extensively on the universality of science and the necessity of providing empirical proof and evidence for scientific theories through experimentation which he himself did.

Although he contributed to multiple disciplines, his greatest achievement was establishing modern optics in the "Book of Optics" and correctly explaining for the first time in history how the process of vision occurs which he proved through his experiments

2. Mohammad Ibn Musa Al-Khawarizmi (780 - 850)

The Persian pioneer developed the discipline of Algebra in "The Compendious Book on Calculation by Completion and Balancing" which took its name from the Arabic word "al-jabr" Khawarizmi used in his book.

The mathematician also introduced the number zero and developed the Hindu-Arabic numeral system which is the numeral system used all over the world today

3. Jabir Ibn Hayyan (721 - 815)

The Persian pioneer, who is considered the most important alchemist in history, established the principles of modern chemistry when he introduced the scientific method to alchemy, the practice that gets its name from the Arabic word "al-kemya'a" which now means chemistry.

Through laboratory experimentation unheard of at his time, Hayyan developed and described the chemical processes crystallization, distillation

and the basic substances that became the foundation of modern chemistry, such as acetic acid, mercury and sulfur.

4. Al-Zahrawi (936 - 1013)

The Arab pioneer **established the foundation of modern surgery** which was enormously influential in the West, many of his innovative surgical instruments and techniques are still used to this day.

Zahrawi invented the **syringe, the forceps, the surgical hook and needle, the bone saw and the lithotomy scalpel**. He was the first physician to describe ectopic pregnancy and the first to identify the genetic nature of haemophilia. He also wrote a thirty-volume encyclopedia of medical practices called "Kitab Al-Tasrif."

5. Modern Medicine - Ibn Sina (980 -1037)

Although the terms "Western medicine" and "modern medicine" are synonymous today, it was the Persian scientist who established the principles of modern medicine.

In his revolutionary book "The Canon of Medicine," Sina introduced advanced drug designing methods considered by scholars to be ages ahead of his time, his techniques, practices and ideas hugely contributed to what is now known as Western medicine.

6. Abu Al-Rayhan Al-Biruni (973 -1048)

The Persian pioneer established techniques to measure the Earth and its distances using triangulation that contributed enormously to modern geography and geodesy.

He solved complex geodesic equations and estimated the radius of the Earth at 6,339.6 kilometers (less than 50 kilometers short of the modern value), a value that was not obtained in the West until the 16th century.